

Diffusive IPS

Motivations

Interacting particles systems are non-equilibrium toy models. Solvable models allow to derive macroscopic law and thermodynamic relations.

- of course poorly realistic as microscopic dynamics (mesoscale).
- on the other side to study macroscopic behavior a lot of microscopic detailed doesn't count: macroscopic fluctuation theory (MFT, kind of generalized Freidlin-Wentzell theory) that describe the stationary states is purely macroscopic.
- non-stochastic derivations from deterministic (in case newtonian) models do very strong assumptions too.
- is know that dynamical systems can be rewritten as markov chains.

Models

Consider the lattice V_N with periodic b.c., i.e. $V_N := \mathbb{T}_N^d$, of mesh 1.
Particle configuration $\eta_t \in \Sigma^{V_N} =: \Sigma_N$, $\Sigma = \{0, 1\}$.

Exclusion process: Markov dynamics on oriented edges E_N (n.n.),

$$\mathcal{L}_N f(\eta) = N^2 \sum_{(x,y) \in E_N} c_{x,y}(\eta) (f(\eta^{x,y}) - f(\eta)), \quad \eta^{x,y}(z) := \begin{cases} \eta(x) - 1 & \text{if } z = x \\ \eta(y) + 1 & \text{if } z = y \\ \eta(z) & \text{if } z \neq x, y \end{cases}. \quad (1)$$

Define the translations $\tau_x \eta(y) := \eta(y - x)$ and $\tau_x h(\eta) := h(\tau_{-x} \eta)$,
We work with *translational covariant* + *reversible* models.

$$\begin{cases} c_{x, x \pm e^{(i)}}(\eta) = \tau_x c_{0, \pm e^{(i)}}(\eta), \text{ e.g. } \eta(x)(1 - \eta(x \pm e^{(i)})) = \tau_x[\eta(0)(1 - \eta(\pm e^{(i)}))], \\ \langle f, \mathcal{L}_N g \rangle_{\mu_N^\rho} = \langle \mathcal{L}_N f, g \rangle_{\mu_N^\rho}, \text{ e.g. } \mu_N^\rho(\eta) = \prod_{x \in V_N} \rho^{\eta(x)} (1 - \rho)^{(1 - \eta(x))} \end{cases}. \quad (2)$$

St. meas. is parametrized by the density ρ : $\mathbb{E}_{\mu_N^\rho}(\eta(x)) := \mathbb{E}_{\mu_N^\rho}(\eta(x)) = \rho$.

Hydrodynamics

The macroscopic evolution of the mass (energy or charge) is related to the *empirical measure* associated to an initial profile $\gamma : \mathbb{T}^d \rightarrow [0, 1]$,

$$\pi_N^t(\eta) := \frac{1}{N^d} \sum_{x \in V_N} \eta_t(x) \delta_x, \quad \int_{\mathbb{T}^d} f d\pi_N(\eta) \xrightarrow[N \rightarrow \infty]{\mathbb{P}_N} \int_{\mathbb{T}^d} f(u) \gamma(u) du.$$

$\pi_N^t(\eta)$ is a cadlag trajectory in $D([0, T], \mathcal{M}^1(\mathbb{T}^d))$ and $\mathbb{P}_N := P_N^\gamma \circ \pi_N^{-1}$.

$$\int_{\mathbb{T}^d} f d\pi_N^t(\eta) \xrightarrow[N \rightarrow \infty]{\mathbb{P}_N} \int_{\mathbb{T}^d} f(u) \rho(u, t) dx, \quad \text{with } \rho \text{ solving } \begin{cases} \partial_t \rho = \nabla \cdot (-J(\rho)) \\ \rho(u, 0) = \gamma(u). \end{cases}$$

$J(\rho) = -D(\rho) \nabla \rho$, *Fick's law*, $D(\rho)$ positive diagonal symmetric matrix .
(3)

We know this with explicit transport coefficients for *gradient systems*.

Gradient models and Diffusive Scaling limits

The *instantaneous current* for our particle models is defined as

$$j_\eta(x, y) := c_{x,y}(\eta) - c_{y,x}(\eta), \text{ note that } j_\eta(x, y) = -j_\eta(y, x). \quad (4)$$

$\mathcal{N}_t(x, y)$:= number of particles that jumped from site x to site y up to time t . The *current flow* across the bond (x, y) is

$$J_t(x, y) := \mathcal{N}_t(x, y) - \mathcal{N}_t(y, x), \text{ note that } J_t(x, y) = -J_t(y, x). \quad (5)$$

Key observation: $M_t(x, y) = J_t(x, y) - \int_0^t j_{\eta(s)}(x, y) ds$ is a martingale.

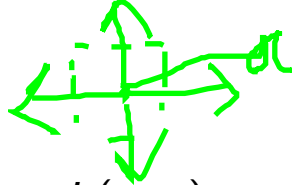
The model is *gradient* when for a local h ($h(\eta) = h(\eta_B)$, B finite $\subset V_N$)

$$j_\eta(x, y) = \tau_y h(\eta) - \tau_x h(\eta), \quad (6)$$

e.g. $j_\eta(x, y) = \eta(x) - \eta(y)$ and $h(\eta) = -\eta(0)$.

Heuristic 1

Discrete mass-continuity equation

$$\eta_t(x) - \eta_0(x) = -\nabla \cdot J_t(x), \quad \nabla \cdot J_t(x) := \sum_{y: \{x,y\} \in \mathcal{E}_N} J_t(x,y). \quad (7)$$


Multiplying by a test function f , dividing by N^d and summing over x

$$\int_{\mathbb{T}^d} f d\pi_N^t(\eta) - \int_{\mathbb{T}^d} f d\pi_N^0(\eta) = -N^{2-d} \int_0^t ds \sum_{x \in V_N} \nabla \cdot j_{\eta_s}(x) f(x) ds + o(1). \quad (8)$$

with discrete integrations by part

$$\int_{\mathbb{T}^d} f d\pi_N^t(\eta) - \int_{\mathbb{T}^d} f d\pi_N^0(\eta) = N^{2-d} \int_0^t ds \sum_{x \in V_N} \tau_x h(\eta_s) \Delta_N f(x) ds + o(1). \quad (9)$$

Heuristic 2

We remind that $\pi_N^t(\eta) \approx \pi^t(du) = \rho(u, t)du$.

When $j_\eta(x, y)$ is gradient we show for a limit point $\mathbb{P}$ of $\{\mathbb{P}_N\}_N$ that

$$\int_{\mathbb{T}^d} f d\pi_N^t(\eta) \xrightarrow[N \rightarrow \infty]{\mathbb{P}_N} \int_{\mathbb{T}^d} f(x) \rho(u, t) du, \quad \begin{cases} \partial_t \rho = \nabla \cdot (-J(\rho)) \\ \rho(u, 0) = \gamma(u). \end{cases}$$

after the replacement of $\tau_x(h(\eta))$ with $\mathbb{E}_{\rho(x/N,t)}(h(\eta))$ in equation (9),

$$\int_{\mathbb{T}^d} f d\pi_N^t(\eta) - \int_{\mathbb{T}^d} f d\pi_N^0(\eta) = N^{2-d} \int_0^t \sum_{x \in V_N} \mathbb{E}_{\rho(x/N, t)}(h(\eta)) \Delta_N \psi(x) ds + o(1).$$

$$J(\rho) = -D(\rho)\nabla\rho, \text{ with } D(\rho) = \frac{da}{d\rho}(\rho) \text{ and } a(\rho(x/N, t)) = \mathbb{E}_{\rho(x/N, t)}(h(\eta)).$$

Reminder

Last time , for SSEP, we got (9) from Dynkin's formula

$$M(t) = F(\eta_t) - F(\eta_0) - \int_0^t dr \mathcal{L}_N F(\eta_r), \quad (10)$$

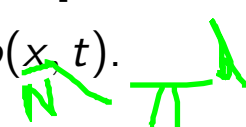
where $F(\eta_t) = \langle \pi_N^t(\eta), f \rangle = \int_{\mathbb{T}^d} f d\pi_N^t(\eta)$ and $h(\eta) = \eta(x)$, therefore (9) was closed in terms of the empirical measure.

Usually in all math papers you see the derivation of the discrete from of the hydrodynamics in this way. But it is not intuitive ("physical " as we did).

Local equilibrium property

This is done by proving the validity of local equilibrium property. This is stated through a replacement lemma which shows that in (probability)

$$\frac{1}{N} \sum_{x \in V_N} \int_0^t \tau_x h(\eta_s) \Delta f(x) ds \simeq \frac{1}{N} \sum_{x \in V_N} \int_0^t a \left(\frac{\int_{B_x} d\pi_N^s(\eta)}{|B_x|} \right) \Delta f(x) ds \quad (11)$$

where B_x is a *microscopically large but macroscopically small box* around the point $x \in V_N$. In 1d imagine $B_x = [x - \varepsilon N, \dots, x, \dots, x + \varepsilon N]$, with " $\varepsilon \ll 1$ and $N \gg 1$ " tuned properly such that $\frac{\int_{B_x} d\pi_N(\xi_s)}{|B_x|} \approx \rho(x, t)$. 

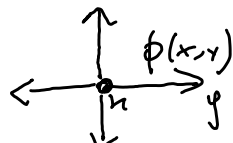
Remark: since $\frac{\int_{B_x} d\pi_N(\xi_s)}{|B_x|} = \frac{1}{|B_x|} \langle \pi_N^t(\eta), 1_{B_x} \rangle$ we close the hydrodynamics in terms of the empirical measure.

Discrete Hodge decomposition 1

Consider the 2d discrete torus and call $V_N, E_N, \mathcal{E}_N$ its vertices, oriented edges and non-oriented one.

A *discrete vector field* ϕ is a map $\phi : E_N \rightarrow \mathbb{R}$ s. t. $\phi(x, y) = -\phi(y, x)$.
Of *gradient type* if $\exists h : V_N \rightarrow \mathbb{R}$ s.t. $\phi(x, y) = [\nabla h](x, y) := h(y) - h(x)$.

The divergence of a discrete vector field ϕ at $x \in V_N$ is defined by



$$\nabla \cdot \phi(x) := \sum_{y : \{x, y\} \in \mathcal{E}_N} \phi(x, y). \quad (12)$$

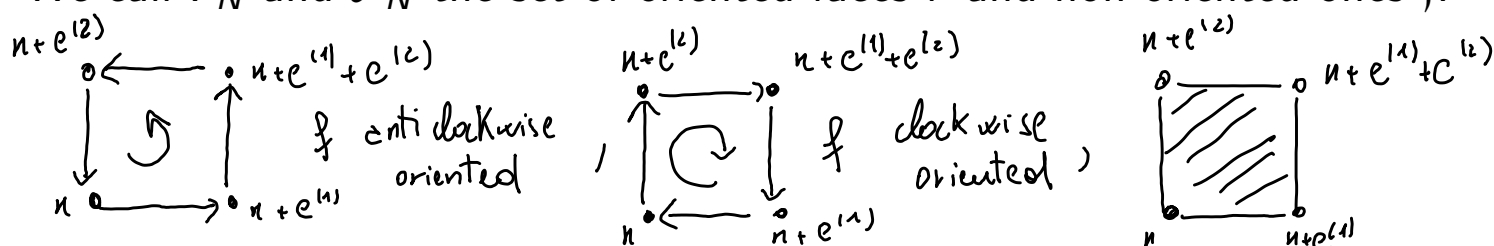
We call Λ^1 the $|\mathcal{E}_N|$ -dimensional vector space of discrete vector fields with

$$\langle \phi, \psi \rangle := \frac{1}{2} \sum_{(x, y) \in E_N} \phi(x, y) \psi(x, y), \quad \phi, \psi \in \Lambda^1. \quad (13)$$

We call Λ^0 the collection of real valued function $\Lambda^0 := \{g : V_N \rightarrow \mathbb{R}\}$.

Discrete Hodge decomposition 2

We call F_N and $\mathcal{F}_N$ the set of oriented faces f and non-oriented ones f .



A 2-form is an antisymmetric map ψ from the set of oriented faces F_N to $\mathbb{R}$, i.e. $\psi(-f) = -\psi(f)$. The boundary operator $\delta : \Lambda^2 \rightarrow \Lambda^1$ is defined by

$$\delta\psi(e) := \sum_{f:e \in f} \psi(f) \cdot \begin{pmatrix} = \psi(f_1) + \psi(-f_2) \\ = \psi(f_1) - \psi(f_2) \end{pmatrix} \quad (14)$$

Λ^2 the vector space of 2-forms $\Lambda^2 := \{\psi : F_N \rightarrow \mathbb{R}\}$.

Remark: (14) is a discrete orthogonal gradient, $\nabla^\perp f$ of a smooth function f is defined as $(-\partial_y f, \partial_x f)$. In higher dimension this a discrete curl.

Discrete Hodge decomposition 3

Remark : By construction $\nabla \cdot \delta\psi = 0$ for any ψ .

$$\nabla \cdot \delta\psi(x) = \sum_{\gamma: (x,y) \in E_N} \delta\psi(x,y) =$$

$[\psi(x) + \psi(-x) = 0]$

The 2-d *discrete Hodge decomposition* is written as the direct sum

$$\Lambda^1 = \nabla\Lambda^0 \oplus \delta\Lambda^2 \oplus \Lambda_H^1, \quad (15)$$

where the orthogonality is with respect to the scalar product (13).

$\nabla\Lambda^0$ are the *gradient fields*. The dimension of $\nabla\Lambda^0$ is $N^2 - 1$. $\delta\Lambda^2$ contains all the vector fields (*circulations*) that can be obtained by (14) from a 2-form ψ . The dimension of $\delta\Lambda^2$ is $N^2 - 1$. The dimension of Λ_H^1 is simply 2. Discrete vector fields in Λ_H^1 are called *harmonic*.

Discrete Hodge decomposition 4

A basis in Λ_H^1 is given by the vector fields $\varphi^{(1)}$ and $\varphi^{(2)}$ defined by

$$\varphi^{(i)}(x, x + e^{(j)}) \stackrel{\circledast}{=} \delta_{i,j}, \quad i, j = 1, 2. \quad (16)$$

If $\varphi \in \Lambda_H^1$ then both $\delta\varphi(x) = (\nabla \cdot \varphi)(x) = 0$ and $d\varphi(f) = 0$, where $d\varphi(f) = \sum_{e:f \ni e} \varphi(e)$.



Given a vector field $\phi \in \Lambda^1$, we write $\phi = \phi^\nabla + \phi^\delta + \phi^H$ to denote the unique splitting in the three orthogonal components.

$$\circledast \quad \varphi^{(1)} = \begin{array}{cccc} \xrightarrow{1} & \xrightarrow{1} & \xrightarrow{1} & \xrightarrow{1} \\ 0 \downarrow & 0 \downarrow & 0 \downarrow & 0 \downarrow \\ \xrightarrow{1} & \xrightarrow{1} & \xrightarrow{1} & \xrightarrow{1} \\ 0 \downarrow & 0 \downarrow & 0 \downarrow & 0 \downarrow \\ \xrightarrow{1} & \xrightarrow{1} & \xrightarrow{1} & \xrightarrow{1} \end{array}, \quad \varphi^{(2)} = \begin{array}{cccc} \uparrow 1 & \uparrow 0 & \uparrow 1 & \uparrow 1 \\ 1 \uparrow & 0 \uparrow & 1 \uparrow & 0 \uparrow \\ \uparrow 1 & \uparrow 0 & \uparrow 1 & \uparrow 1 \\ 1 \uparrow & 0 \uparrow & 1 \uparrow & 0 \uparrow \end{array}$$

The non-gradient problem

In physics the critical dimension is $d = 2$ and the first one to be realistic.

In dimension greater than 1, reversible gradient models are very rare.

Conjecture:

In dimension greater than one the only one irreducible exclusion process to be reversible and gradient is the symmetric simple exclusion process.

Also in hydrodynamics of IPS physics changes in dimension greater than one, we will the appearance of vortices for example.

Back to the microscopic current

We already pointed out that $j_\eta(x, y) = -j_\eta(y, x)$, i.e. discrete vector field.

We know then

$$j_\eta(x, y) = j_\eta^\nabla(x, y) + j_\eta^\delta(x, y) + j_\eta^H(x, y). \quad (17)$$

Remark: here we have a functional dependence on the configuration η , moreover from $c_{x, x \pm e^{(i)}}(\eta) = \tau_x c_{0, \pm e^{(i)}}(\eta)$ we have also

$$j_\eta(x, x \pm e^{(i)}) = \tau_x j_\eta(0, \pm e^{(i)}). \quad (18)$$

e.g. $j_\eta(x, x + e^{(i)}) = \eta(x) - \eta(x + e^{(i)}) = \tau_x(\eta(0) - \eta(e^{(i)}))$.

From (17) and (18) $\rightarrow$ the following functional Hodge decomposition.

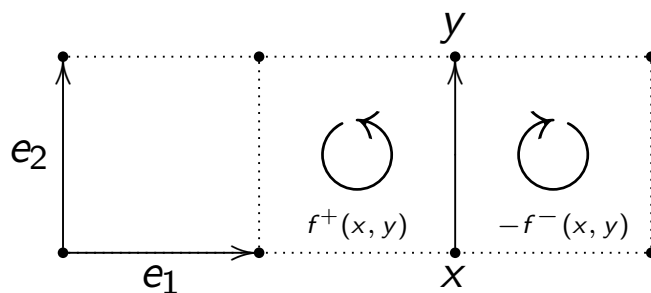
Functional Hodge decomposition (2-d case) 1

Theorem

Let $j_\eta(x, y)$ be translation covariant, we have

$$j_\eta(x, y) = [\tau_y h(\eta) - \tau_x h(\eta)] + [\tau_{\mathfrak{f}^+(x, y)} g(\eta) - \tau_{\mathfrak{f}^-(x, y)} g(\eta)] \pm C^{(i)}(\eta). \quad (19)$$

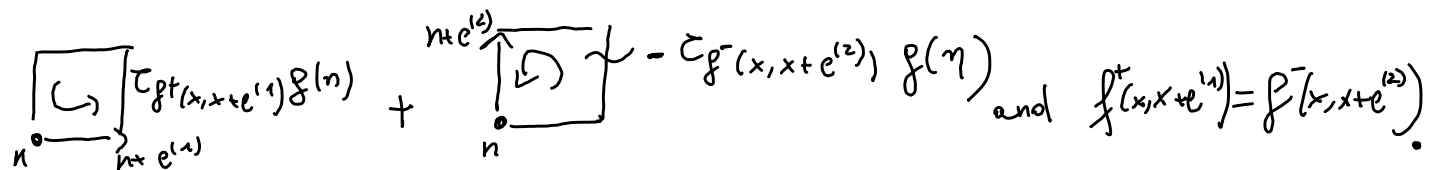
$C^{(i)}(\eta) = \frac{1}{N^2} \sum_{x \in V_N} j_\eta(x, x + e^{(i)})$ are translational invariant and unique. Where $\tau_{\mathfrak{f}} := \tau_x$ with $\mathfrak{f}$ centered in $x + \frac{e_1}{2} + \frac{e_2}{2}$.



Functional Hodge decomposition (2-d case) 2

$$j_{\eta}^{\nabla}(x, y) = [\tau_y h(\eta) - \tau_x h(\eta)], \quad j_{\eta}^{\delta}(x, y) = [\tau_{\hat{f}^+(x, y)} g(\eta) - \tau_{\hat{f}^-(x, y)} g(\eta)] \text{ and } j_{\eta}^H(x, x + e^{(i)}) = C^{(i)}(\eta).$$

Remark: $\operatorname{div} j_{\eta}(x) = \operatorname{div} j_{\eta}^{\nabla}(x)$, the parts of g and $C^{(i)}$ are divergence free.



$$\tau_{\hat{f}^+(x, x+e^{(1)})} g(\eta) + \tau_{\hat{f}^-(x, x+e^{(2)})} g(\eta) \text{ and } \hat{f}^+(x, x+e^{(1)}) = \hat{f}^-(x, x+e^{(2)})$$

When we have a decomposition with h and g local

$$j_{\eta}(x, y) = [\tau_y h(\eta) - \tau_x h(\eta)] + [\tau_{\hat{f}^+(x, y)} g(\eta) - \tau_{\hat{f}^-(x, y)} g(\eta)], \quad (20)$$

$j_{\eta}^{\delta}(x, y)$ will be interpreted as a current of "microscopic vortices" that generates macroscopic vortices.